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Behavioral despair is differentially affected by the length and timing of photic stimulation in the dark phase of an L/D cycle

D. Schulz [1], A. Aksoy [2], R. Canbeyli [*,2]

Psychobiology Laboratory, Boğaziçi University, Istanbul, Turkey

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ABSTRACT

The effect of varying the length and timing of photic stimulation in the dark phase of an L/D lighting cycle on behavioral despair was investigated in female Wistar rats. Animals were kept in a vivarium on an L/D 12 h:12 h light cycle (lights on at 0700 h) except for a single day of light exposure in an insulated chamber in the dark phase of the L/D schedule. Light pulses provided by an incandescent lamp (15- and 25-W, for Exps. 1 and 2, respectively) either 2-h (Exp. 1) or 30-min in length (Exp. 2) were administered to independent groups of rats (n=8 each) either in the early, middle or late hours of the dark phase of the L/D cycle in the insulated chamber. Light pulses were delivered beginning 2 1/2, 5 1/2 or 7 1/2 h (Exp. 1) or 3 1/4, 6 1/4 and 8 1/4 h (Exp. 2) after dark onset. Control animals were treated similarly except for photic stimulation. In each experiment, an additional group received a light pulse of the appropriate length both in the early and late portion of the dark phase (double double-pulse groups): beginning 2 1/2 and 7 1/2 h (Exp. 1) or 3 1/4 and 8 1/4 h (Exp. 2) after dark onset. All animals then underwent two forced swim tests separated by 24 h with the first test occurring in the light (starting at 1500 h) following the dark phase when photic stimulation was administered. Total duration of immobility in the second swim test was measured to gauge behavioral despair. In Exp. 1, the 2-h double double-pulse group showed significantly shorter immobility compared to controls (p<0.05). In Exp. 2, 30-min light pulse delivered late in the dark phase reduced immobility significantly compared to controls and all the other light-treated groups (p<0.01). Results indicate that photic stimulation may have antidepressant effect on behavioral despair depending on the timing and the duration of photic stimulation.

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1. Introduction

Light therapy, treatment of patients suffering from depression with bright light exposure, has been used successfully to alleviate symptoms of both seasonal (Avery, 1998; Goodwin et al., 1982; Lewy et al., 1982, 1987; Oren and Terman, 1998; Rosenthal et al., 1984; Rosenthal and Wehr, 1992; Wehr and Wirz-Justice, 1982) and nonseasonal (Kripke, 1998; Kripke et al., 1983, 1992; Mackert et al., 1991) depression. The clinical insight on depression gained by light therapy may benefit from animal studies that can provide the basis for elucidating the ameliorative effect of light treatment on the central mechanisms of depression. There is, however, a paucity of studies that have directly tested the effect of light treatment on depression in animals. A study by Molina-Hernandez and Tellez-Alcantara (2000) showed that prolonged exposure of male rats to a long photoperiod (14 h:10 h L/D cycle) has an antidepressant effect as measured by reduced immobility in a forced swim test. We recently reported that in female Wistar rats a single 12 h exposure to light in the dark phase of an L/D (12 h:12 h) cycle provides immunity to behavioral despair (Yilmaz et al., 2004), an animal model of depression (Porsolt et al., 1977, 1978). The model is based on two forced swim tests separated by 24 h. Many antidepressants used to treat depression in humans decrease whereas compounds with depressive effects in humans increase immobility in the second swim test compared to controls (Dalvi and Lucki, 1999; Lucki, 1997; Porsolt et al., 1977, 1978; Reneric and Lucki, 1998).

Our finding that a 12-h exposure to light in the dark phase of an L/D cycle attenuates the effect of behavioral despair suggests the possibility that an antidepressant effect may be attained by shorter light pulses. Probing the dark phase of the L/D cycle with short light pulses in an animal model of depression may provide the basis for the investigation of central mechanisms mediating the antidepressant effect of light. This is especially critical since light therapy in clinical applications has been used with varying degrees of success with different durations and timing of photic stimulation with respect to

Abbreviations: HPA, Hypothalamo-Pituitary-Adrenal Axis; L/D, Light/Dark Cycle; SCN, Suprachaiasmatic Nucleus; PRC, Phase Response Curve.

* Corresponding author. Department of Psychology, Bogazici University, 80815 Bebek, Istanbul, Turkey. Tel.: +90 212 359 7129; fax: +90 212 287 2472.
E-mail address: canbeyli@boun.edu.tr (R. Canbeyli).
[1] Present address: Institute of Physiological Psychology, University of Duesseldorf, Universitaetstrasse 1, D-40225 Düsseldorf, Germany.
[2] Present address: Psychobiology Laboratory, Department of Psychology, Bogazici University, 34342 Bebek, Istanbul, Turkey.

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the circadian rhythms (Kripke, 1991; Lewy et al., 1998; Partonen, 1994; Wehr et al., 1986). A major target for the antidepressant effect of light may be the suprachiasmatic nucleus (SCN), the master pacemaker of the brain that regulates a large number of circadian rhythms (Turek, 1985, 1994; van Esseveldt et al., 2000). The SCN is implicated in affective functions and disorders (Buijs et al., 1997, 1998, 1999; Engelmann et al., 1998; Van Cauter and Turek, 1986). Photic stimulation alters the phase of biological rhythms via the SCN: light pulses delivered early or late in the dark phase of an L/D cycle cause, respectively, phase delays or advances of biological rhythms (Honma et al., 1978; Khalsa et al., 2003; Moore-Ede et al., 1982; Summer et al., 1984). The present study therefore explored the ameliorative effect of light exposure in female Wistar rats on behavioral despair by varying the length (2 h and 30 min) and the timing of single light pulses in the dark phase of a 12 h:12 h L/D cycle. A pair of light pulses administered in the early and late stage of the dark phase was also utilized to determine whether the effect of the pulses is cumulative.

Finally, as in the Yilmaz et al. study (2004), the present study also aimed to investigate the effect of light treatment on female rats considering the fact that gender affects the incidence and severity of depression in humans (Bracke, 1998; Dam et al., 1998; Lee and Chan, 1998; Nilssen et al., 1999) and animals (Barros and Ferigolo, 1998; Brotto et al., 2000; Kennett et al., 1986).

2. Materials and methods

2.1. Subjects

A total of 72 female Wistar rats from our breeding colony, weighing 190–230 g, were used in the experiment. Subjects were group housed in plastic cages with food and water available ad libitum. They were maintained on a 12 h:12 h L/D cycle (lights on at 0700 h) in a temperature controlled vivarium (22±2 °C) except for a single day of light treatment in an insulated chamber.

2.2. Apparatus

Light treatment was administered in a ventilated chamber with sound and light insulation. Light was provided by a tungsten lamp approximately 50 cm above the cage housing the subjects.

2.3. Experimental procedure

The study was conducted in two phases: in the first experiment, independent groups of rats were exposed to 2 h pulses of light (experimental groups) or were kept in the dark (controls) in the dark phase of the L/D cycle. In the second experiment, independent groups of rats were administered 30 min pulses of light at different times in the dark phase. All animals in the two experiments were subjected to two forced swim tests separated by 24 h.

For light treatment, subjects were taken from their home cages 10 min before the beginning of the dark period in the L/D cycle (at 1850 h) and introduced into the insulated chamber in pairs, each pair housed in a plastic cage (two pairs in two cages were run on any given day). All pairs were formed from among cage mates. Animals had free access to food and water in the chamber. The chamber was kept dark in the 12 h dark phase of the L/D cycle except for light exposure(s) for different groups as noted below. For all groups, light was turned on at 0700 h in the insulated chamber to start the light phase of the L/D cycle. Animals were kept in the insulated chamber until 1500 h when the first of the swim test was conducted as specified below. Animals were returned to their home cages in the vivarium after the first test and were maintained on their usual 12L:12 D light cycle.

2.4. Forced swim tests

Animals were tested individually in two forced swim tests. Following common practice in the literature, the first swim test lasted 15 min, followed 24 h later by a 5 min second test. Swim tests were conducted in a Plexiglas cylinder of 45 cm height and 30 cm diameter filled with fresh water (25°) to a height of 15 cm. After each test, the animal was towel dried and placed under a lamp for 30 min for drying.

2.5. Measurements and statistical treatments

Swim tests were recorded on videotape and total duration of immobility in the second test was measured. Immobility was defined as floating or remaining motionless without leaning against the wall of the cylinder. Analysis of variance was used to assess the effects of treatment followed by Newman–Keuls tests to assess significant group effects ($p<0.05$ or better).

2.6. Experiment 1

Independent groups of rats ($n=8$ each) were exposed to light (15-W; approximately 105 lx inside the cage at the level of the animal) for 2 h beginning either 2 1/2, 5 1/2, or 7 1/2 h ('early', 'middle' and 'late' long-pulse groups) after dark onset. Another group received two exposures of photic stimulation of 2 h each beginning 2 1/2 and 7 1/2 h after dark onset ('2-h double-pulse group'). Animals in the control group ($n=8$) were placed in the chamber and treated the same way as the experimental animals except that they were not exposed to light.

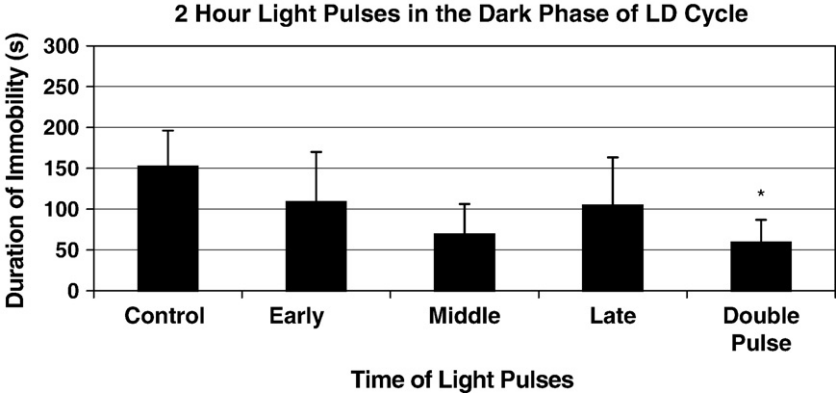


Fig. 1. Duration of immobility (mean±SD) in the second of two swim tests separated by 24 h. Independent groups of rats ($n=8$) were exposed to light (15- WW) for 2 h beginning either 2 1/2, 5 1/2 or 7 1/2 h ('early', 'middle' and 'late' groups) after dark onset. The double-pulse (DP) group received two light pulses each 2 h long starting 2 1/2 h and 7 1/2 h after dark onset. Control animals ($n=8$) were placed in the chamber and treated the same way as the experimental groups except that they were not exposed to light. Asterisk indicates that the double-pulse group was significantly different from controls ($p<0.05$).

2.7. Experiment 2

A total of 32 female Wistar rats were used in the second phase of the experiment with experimental conditions identical to the first except for the duration, intensity and timing of the light treatment. Because of the shorter duration of photic stimulation in the second experiment (30 min), brighter light (provided by a 25-W lamp compared to the 15-W lamp in the first experiment) was used. Four groups (n=8 each) received 30-min long light in such a way that the midpoint of the pulse corresponded to that of the appropriate 2 h pulse in Exp. 1. Thus three groups received pulses beginning 3 1/4, 6 1/4, or 8 1/4 h ('early', 'middle' and 'late' night short-pulse groups) after dark onset, while another group received two pulses starting 3 1/4 and 8 1/4 h after dark onset ('30 min double double-pulse group').

3. Results

3.1. Experiment 1

Total duration of immobility (mean±SD) in the second swim test is shown for all the groups in Fig. 1. One-way ANOVA indicated a significant group difference [F (4,35)=3.00, p<0.05]. Post hoc tests showed that the double-pulse group displayed significantly shorter duration of immobility compared to controls (p<0.05). There were no significant differences in the duration of immobility among the other groups. Because the middle pulse group also showed considerably reduced immobility compared to the controls, a Dunnett test was used to compare the controls with each of the light pulse groups. This test indicated a significant difference between the control group and the double pulse as well as the middle pulse groups (p<0.05).

The results indicate that light exposure is significantly effective only in the double-pulse group which received longer light treatment than the other groups. Compared to controls, the middle pulse group also showed considerable reduction in immobility as indicated by the Dunnett (but not Newman–Keuls) tests. Our previous study (Yilmaz et al., 2004) indicated that a 12-h long light treatment in the dark phase of an L/D cycle has ameliorative effect on behavioral despair. The present findings indicate that a shorter light exposure totaling 4 h in the dark phase of a similar schedule is effective in reducing behavioral despair. The next study aimed to further refine the timing of the light treatments by reducing the treatment duration to 30 min. Because the intensity of light may matter, groups were exposed to light from a 25 W incandescent lamp (approximately 195 lx inside the cage at the level of the animal) instead of the 15 W lamp used in the first experiment. Thus, while the duration of light exposure was reduced to a quarter of the previous experiment, brightness was approximately doubled.

3.2. Experiment 2

Fig. 2 shows the duration of immobility (mean±SD) in the two swim tests for the four light-treated groups in this phase of the experiment together with the results of the control group from Exp. 1. One-way ANOVA for immobility using the data for the four light-treated groups and the controls from Exp. 1 indicated significant group difference [F (4, 35)=5.20, p<0.01]. Post hoc tests showed that the late pulse group that received light treatment starting 8 1/4 h after dark onset had significantly shorter duration of immobility than the control group and all the other light-treated groups (all p<0.01). There were no other significant differences.

The results indicate that a single 30 min short pulse of light delivered in the late aspect of the dark phase in an L/D cycle has ameliorative effect on behavioral despair. As in the first study, pulses delivered either early or the middle portion of the dark phase did not have an ameliorative effect on forced swimming.

4. Discussion

The present findings suggest that a single, brief exposure to light treatment in the dark phase of a 12:12 L/D lighting schedule can have an antidepressant effect on behavioral despair as measured by decreased duration of immobility in the second of two forced swim tests compared to controls. Previously, Molina-Hernandez and Tellez-Alcantara (2000) had reported that a 30-day exposure to a long day (L/D 14:10 cycle) has a protective effect on forced swimming. More recently, we showed that a single 12-h exposure to light in the dark phase of an L/D cycle has protective effect on behavioral despair. The present findings further refine the results of our previous study and constitute the first report of an ameliorative effect of a single, short light pulse treatment on forced swimming.

The positive effect of light treatment on behavioral despair seems to be dependent on both the timing and duration of immobility. Single light pulses either 30-min or 2-h long delivered in the early or middle portion of the dark phase did not have any significant ameliorative effect on forced swimming, whereas a single 30-min pulse delivered late in the dark phase of the L/D cycle proved the most efficient in immunizing rats against behavioral despair. This finding is noteworthy in that light treatment early in the morning is more effective than treatment in the evening (Kripke et al., 1983; Kripke, 1998; McClung, 2007; Terman and Terman, 2005; Westrin and Lam, 2007).

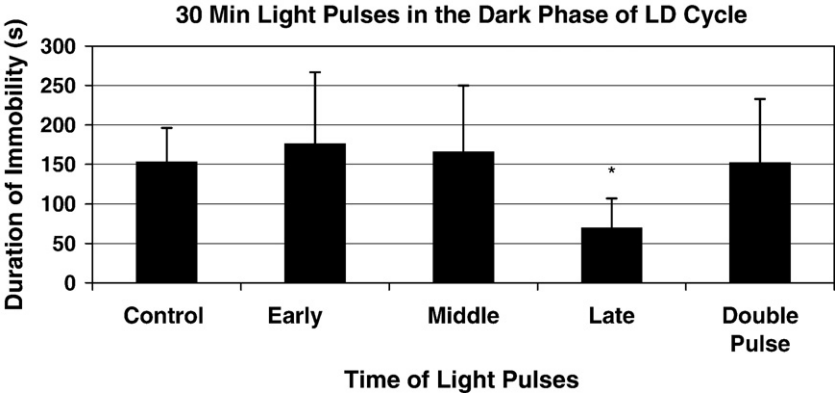


Fig. 2. Duration of immobility (mean±SD) in the second of two swim tests separated by 24 h. Independent groups of rats (n=8) were exposed to light (25-W) for 30 min beginning either 3 1/4, 6 1/4, or 8 1/4 h ('early', 'middle' and 'late' night short-pulse groups) after dark onset while another group received two pulses of 30 min length 3 1/4 and 8 1/4 h after dark onset ('30 min double-pulse group'). The controls, taken from Exp. 1, were treated similarly but not exposed to light. Asterisk indicates that the late pulse group was significantly different from controls and the other light pulse groups (p<0.01 for each comparison).

Since we used different intensities of light in the 2-h and 30-min light exposure groups, we cannot directly compare the ameliorative effect of different durations of light exposure on behavioral despair. Nevertheless, our results indicate that the 2-h photic exposure used in Exp. 1 can also be effective in alleviating behavioral despair, as indicated by reduced immobility in the middle pulse group. In particular, a combination of early and late light treatment of 2 h duration had a significant antidepressant effect on behavioral despair. This is also noteworthy in that a review of light treatment in nonseasonal depression indicated that a combination of morning and evening light treatments works better than either alone (Kripke, 1998).

The mechanism by which light can modulate depression in humans or behavioral despair in rats is not fully understood. Nevertheless, previous findings point to possible mechanisms that may mediate the ameliorative effect of light on behavioral despair. Light treatment, for instance, can have an arousing effect on animals that may subsequently protect them against the stress of forced swim tests. Nonspecific arousal, however, is not able to explain the differential effect of pulses of different lengths and times of occurrence in the dark phase on behavioral despair. Moreover, arousal per se cannot account for the prolonged effect of a single light treatment before the first swim test on the shortened duration of immobility in the second swim test administered 24 h later.

Potentially a more valid explanation of the present results may be that photic stimulation at different times in the dark phase of the L/D cycle may have affected the suprachiasmatic nuclei (SCN) differentially. Light stimulation early in the dark phase delays and late in the dark phase advances the regulation of behavioral and biological rhythms by the SCN (Honma et al., 1978; Moore-Ede et al., 1982; Summer et al., 1984). Even though we did not have the means to assess the potential phase shifts in the circadian rhythms in the present study, it is likely that both the timing and the duration of the pulses interacted with the capacity of light to phase shift the SCN depending on where and for how long it occurs in the Phase Response Curve (PRC) of the biological clock (Honma et al., 1978; Moore-Ede et al., 1982; Pittendrigh and Daan, 1976; Rosenwasser and Dwyer, 2002; Summer et al., 1984; van Esseveldt et al., 2000). Effects of light falling in different aspects of the PRC may explain the differential effects of the middle and double pulses observed in the two experiments. Interestingly, the 2-h but not the 30-min long middle pulse resulted in a near significant antidepressant effect as shown by the Dunnett test. These results suggest further exploration of this segment of the dark phase for ameliorative effect of light pulses. The possibility that the longer middle pulse was more efficacious in inducing a phase shift (in particular a phase advance) than the 30 min middle pulses may be advanced as a reason for the differential result with the two middle pulses. In the same vein, the fact that the 2-h double pulses proved to have antidepressant effect whereas the 30-min long double pulses did not suggests that the impact of photic treatment on behavioral despair is not a simple cumulative effect. This finding should be evaluated in the context of the report that the effect of multiple light pulses in the dark phase of an L/D cycle on the PRC of rats is not a simple cumulative one that can be fully predicted on the basis of phase shifts derived from single pulses (Stokes et al., 1999).

Light is the major input to the SCN, the master pacemaker of the brain that regulates a large number of circadian rhythms including that of melatonin (Moore-Ede et al., 1982; Turek, 1985, 1994; van Esseveldt et al., 2000). Therefore, a phase shift by light can have the dual impact of altering the circadian rhythmicity of melatonin secretion by the pineal gland and that of many other biological functions that are regulated by the SCN (Moore-Ede et al., 1982; van Esseveldt et al., 2000). While reproductively not photoperiodic (Vanecek and Illnerova, 1982; Wallen et al., 1987), Wistar rats nevertheless are affectively photoperiodic in that they display behavioral despair and exploratory anxiety to short day regimens (Prendergast and Kay, 2008) suggesting that they may also be affectively sensitive to modulation of melatonin secretion by light exposure. Both phase shiftings of circadian rhythms mediated by the SCN and direct photic stimulation are capable of disrupting melatonin secretion (Kanematsu et al., 1994; Kennaway and Rowe, 1994; Moore-Ede et al., 1982). An acute disruption of melatonin secretion or its rhythmicity, however, is not likely to explain the present results since administration of melatonin or its agonists has been shown to have an antidepressant effect in rodents as assessed by reduced immobility in forced swim tests (Bourin et al., 2004; Brotto et al., 2000; Overstreet et al., 1998; Raghavendra et al., 2000). In view of these findings, the effect of disrupting melatonin secretion in the present study, if at all, should have been in the direction of aggravating not attenuating behavioral despair.

The ameliorative effect of light pulses in the present study may be due to a chain of events other than melatonin secretion set into motion by the SCN. Recent studies have uncovered a variety of roles for the SCN in affective functions and disorders (Buijs et al., 1997, 1998, 1999; Engelmann et al., 1998; Van Cauter and Turek, 1986). There seems to be a reciprocal relationship between SCN function and stress related behaviors; stress and psychopathology can impair biological rhythms (Kripke et al., 1978; Meerlo et al., 1996; Stewart et al., 1990a; Tornatzky and Miczek, 1993), while disruption of biological rhythms can induce stress (Souetre et al., 1989; Stewart et al., 1990b). Neuroanatomical studies reveal extensive and usually reciprocal relationship between the SCN and several limbic structures critically involved in affective behavior and psychopathology (Buijs et al., 1998; Leak and Moore, 2001; Vrang et al., 1995; Watts and Swanson, 1987; Watts et al., 1987). The neurochemical basis for the SCN involvement in depression may be related to its role in the regulation of the hypothalamo-pituitary-adrenal (HPA) axis response to stress (Buijs et al., 1993, 1998, 1999; Kalsbeek et al., 1992; Sage et al., 2001). Moreover, we recently showed that duration of immobility in the forced swim tests is reduced in male Wistar rats after bilateral destruction of the SCN (Tataroglu et al., 2004).

It is possible that the antidepressant effect of light on behavioral despair may follow a route not directly related to the SCN, namely through a serotonergic mechanism activated by light exposure. Serotonergic activation can modulate behavior in a phase-dependent manner (Mistleberg and Antle, 1998). To explain the present results, another pathway that connects the retinal light input with structures capable of modulating stress and depression via a serotonergic mechanism (Lambert et al., 2002; Maier and Watkins, 2005; Montgomery, 1995) can be considered. For example, in addition to the well-documented pathways connecting the retina to the SCN directly or via the intergeniculate leaflet, recent studies show that a direct path exists in the rat between the retina and the dorsal raphe nuclei (Fite et al., 1999, 2005; Kawano et al., 1996; Shen and Semba, 1994). Serotonergic projections from the dorsal raphe nuclei in turn may modulate the activities of several structures — including the SCN — involved in regulating mood and affective behavior (Dinan, 1996; Glass et al., 2000; Hay-Schmidt et al., 2003; Kennaway and Moyer, 1998; Maier and Watkins, 2005; Meyer-Bernstein and Morin, 1999; Moga and Moore, 1997; Montgomery, 1995).

5. Conclusion

The present findings indicate that a single brief exposure to light in the dark phase of an L/D cycle can provide immunity to behavioral despair in female rats. Furthermore, the ameliorative effect of light is dependent on both its time of occurrence in the dark phase and on its duration. These results necessitate a detailed investigation into the critical time and/or the intensity of light exposure for optimal ameliorative effect of photic treatment and at the same time provide a framework for the investigation of the mechanism whereby light serves as an antidepressant agent in depression. Specifically, future studies can assess the cumulative antidepressant effect of pulses delivered over days as well as the potentially lasting effect of a single

pulse. An important avenue of investigation will be to assess the potential phase shifts as well as alterations in the rhythmicity of activity in response to pulses delivered in the early and late phases of the dark period.

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